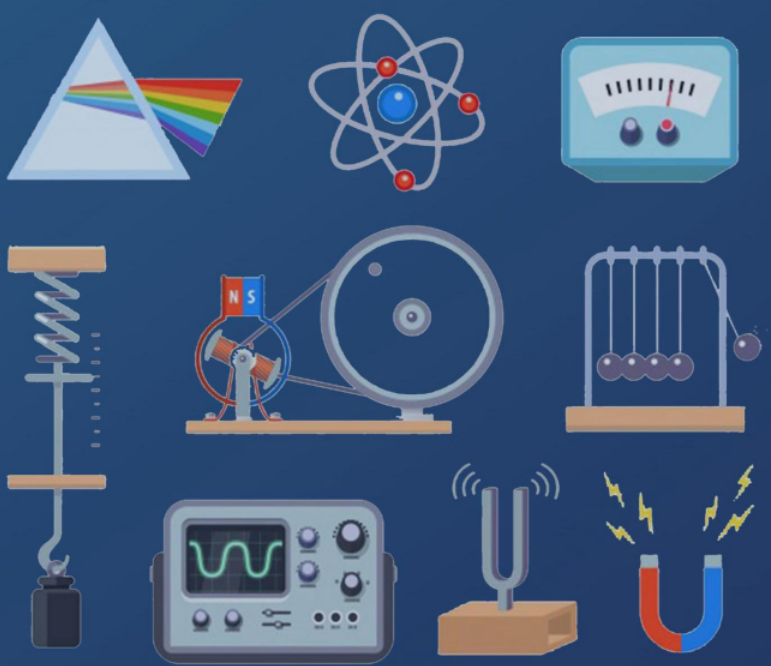
PHYSICS

Friction



SHEET

eSaral.com

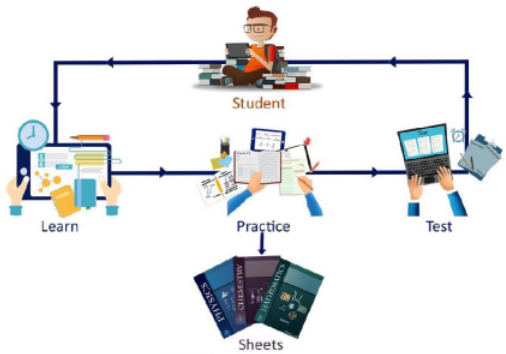
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr. No.	Topic	Ex - 1	Ex - 1A	Ex - 2	Ex - 2A
1	Static and Kinetic Friction	1-5	1		
2	Basic Problems in Friction	6-8	2	1	
3	A Systematic Approach to Friction	9,10	3,5-8	2-4	1,2
4	Two Block Problems	11-14		5,6	3,6-10
5	Angle of Repose	15,16	9.10	7,8	
6	Net Contact Force	17	4		4
7	Miscellaneous Problems			9	5

Exercise - 1 Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Static and Kinetic Friction

Q1. The coefficient of static friction between two surface depend on-

(A) Nature of surface

(B) The shape of the surface in contact

(C) The area of contact

(D) All of the above

Q2. Maximum value of static friction is known as :-

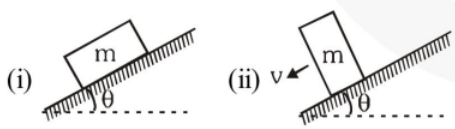
(A) Limiting friction

(B) Kinetic friction

(C) Normal reaction

(D) Coefficient of friction

Q3. A block is first placed on its long side and then on its short side on the same inclined plane, as shown. The block slides down the plane on its short side but remains at rest on its long side. A possible explanation is:



(A) the short side is smoother than long side

(B) the frictional force is less because the contact area is less

(C) the center of gravity is higher in the second case

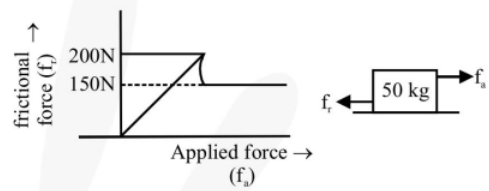
(D) the normal force is less in the second case

Q4. A 20 kg block is resting on a horizontal floor. A force of 75 N is required to just move the block and a force of 60 N is required to move the block with constant velocity then coefficient of static friction is :-

(A) 0.375 (B) 0.44

(C) 0.52 (D) 0.60

Q5. The graph between applied force and frictional force for a body of mass 50 kg placed on a horizontal surface is as shown in figure the value of cofficient of static and kinetic friction will be :

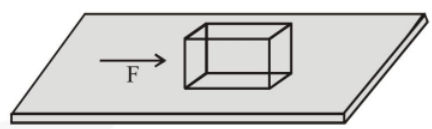


(A) 0.4, 0.3 (B) 0.3, 0.2

(C) 0.4, 0.2 (D) 0.2, 0.3

Basic Problems in Friction

Q6. A 2 kg block is resting on horizontal surface. If coefficient of static friction is 0.4 and 2.5 N force is applied on the block then the friction on the block is :-



(A) 2.5 N (B) 5 N

(C) 7.84 N (D) 10 N

Q7. A block is mass 'm' is resting on the floor of a lift which is moving downward with acceleration of 'g'. If the block is pulled along the surface and coefficient of friction is μ then the friction on block is :-

(A) mg (B) μmg

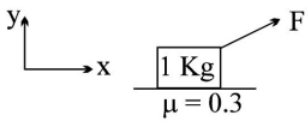
(C) 2 μmg (D) Zero

Q8. A marble block of mass 2 kg lying on ice when given a velocity of 6 m/s is stopped by friction in 10s. Then the coefficient of friction is-

(A) 0.02 (B) 0.03 (C) 0.06 (D) 0.01

A Systematic Approach to Friction

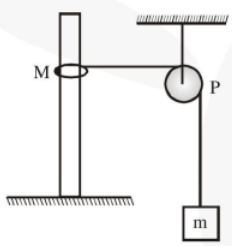
Q9. A force $\vec{F} = \hat{i} + 4\hat{j}$ acts on block shown. The force of friction acting on the block is :



(A) $-\hat{i}$ (B) $-1.8\hat{i}$

(C) $-2.4\hat{i}$ (D) $-3\hat{i}$

Q10. In the figure shown a ring of mass M and a block of mass m are in equilibrium. The string is light and pulley P does not offer any friction and coefficient of friction between pole and M is μ. The frictional force offered by the pole on M is :



(A) Mg directed up

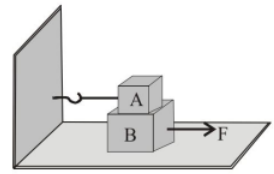
(B) μmg directed up

(C) $(M - m)$g directed down

(D) μmg directed down

Two Block Problems

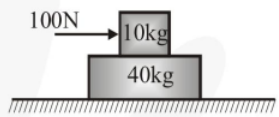
Q11. A block A with mass 100 kg is resting on another block B of mass 200 kg. As shown in fugure a horizontal rope tied to a wall holds it. The coefficient of friction between A and B is 0.2 while coefficient of friction between B and the ground is 0.3. The minimum required force F to start moving B will be:-



(A) 900 N (B) 100 N

(C) 1100 N (D) 1200N

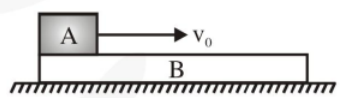
Q12. A 40 kg slab rests on a frictionless floor. A 10 kg block rests on top of the slab, fig. The static coefficient of friction between the block and the slab is 0.60 while the kinetic coefficient is 0.40. The 10 kg block is acted upon by a horizontal force of 100N. If g = 9.8 m/s^2, the resulting acceleration of the slab will be :-



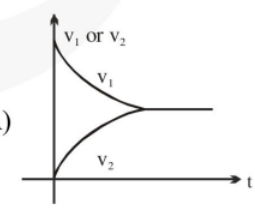
(A) 0.98 m/s^2 (B) 1.47 m/s^2

(C) 1.52 m/s^2 (D) 6.1 m/s^2

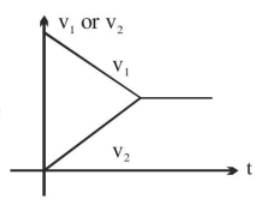
Q13. A block A is placed over a long rough plank B of same mass as shown in figure. The plank is placed over a smooth horizontal surface. At time t = 0, block A is given a velocity v_0 in horizontal direction. Let v_1 and v_2 be the velocities of A and B at time t. Then choose the correct graph between v_1 or v_2 and t.



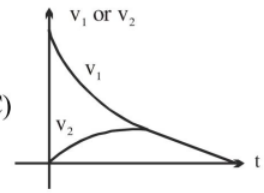
(A)



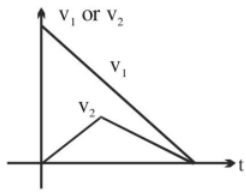
(B)



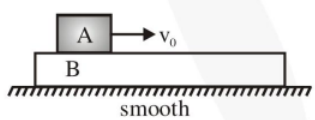
(C)



(D)



Q14. A block A of mass m is placed over a plank B of mass 2 m. Plank B is placed over a smooth horizontal surface. The coefficient of friction between A and B is 0.5. Block A is given a velocity v_0 towards right. Acceleration of B relative to A is :-



(A) $\frac{g}{2}$ (B) g (C) $\frac{3g}{4}$ (D) zero

Angle of Repose

Q15. A block is at rest on an inclined plane making an angle α with the horizontal. As the angle α of the incline is increased, the block starts slipping when the angle of inclination becomes θ. The coefficinet of static friction between the block and the surface of the inclined plane is :-

(A) $\sin\theta$ (B) $\cos\theta$
(C) $\tan\theta$ (D) $\cot\theta$

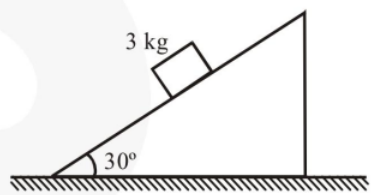
Q16. A block is pushed with some velocity up a rough inclined plane. It stops after ascending few meters and then reverses its direction and returns back to point from where it started. If angle of inclination is 37° and the time to climb up is half of the time to return back then coefficient of friction is :

(A) $\frac{9}{20}$ (B) $\frac{7}{5}$

(C) $\frac{7}{12}$ (D) $\frac{5}{7}$

Net Contact Force

Q17. A block of mass 3 kg is at rest on a rough inclined plane as shown in the figure. The magnitude of net force exerted by the surface on the block will be ($g=10\ m/s^2$)



(A) 26 N (B) 19.5 N
(C) 10 N (D) 30 N

Exercise - 1A — *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Which of the following is/are correct :-

(A) Kinetic friction is self adjustible.

(B) Friction always oppose motion.

(C) Friction may cause motion.

(D) Kinetic Friction opposes relative motion

Q2. A block is kept on a rough horizontal surface as shown. Its mass is 2 kg and coefficient of friction between block and surface (μ) = 0.5. A horizontal force F is acting on the block. When



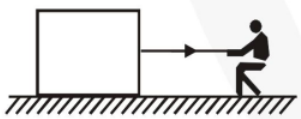
(A) F = 4 N, acceleration is zero.

(B) F = 4 N, friction is 10 N and acceleration is 3 m/s^2.

(C) F = 14 N, acceleration is 2 m/s^2.

(D) F = 14 N, friction is 14 N.

Q3. A man pulls a block heavier than himself with a light rope. The coefficient of friction is the same between the man and the ground and between the block and the ground. Then



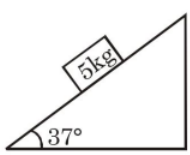
(A) block will never move

(B) the man can move even when the block is stationary

(C) if both move $a_{man} > a_{block}$

(D) none of these

Q4. A block of mass "5kg" is placed on a wedge having inclination of 37° with the horizontal. Coefficient of friction between block and wedge is 0.8. Then select the correct statement(s) :



(A) Friction force between block and the wedge is 32 N.

(B) Friction force between block and the wedge is 30 N.

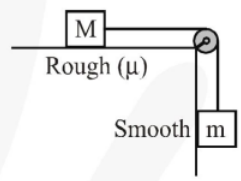
(C) Contact force between block and wedge is 50 N.

(D) Contact force between block and wedge is 40 N.

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

In figure, two blocks M and m are tied together with an inextensible and light string. The mass M is placed on a rough horizontal surface with coefficient of friction μ and the mass m is hanging vertically against a smooth vertical wall. The pulley is frictionless.



Q5. Choose the correct statement(s) :

(A) The system will accelerate for any value of m

(B) The system will accelerate only when m > M

(C) The system will accelerate only when m > μM

(D) Nothing can be said

Q6. Choose the correct statement(s) related to the tension T in the string.

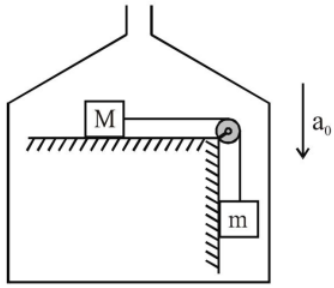
(A) When m < μM, T = mg

(B) When m < μM, T = Mg

(C) When m > μM, μMg < T < mg

(D) When m > μM, mg < T < μMg

Q7. Imagine a situation in which the given arrangement is placed inside an elevator that can move only in the vertical direction and compare the situation with the case when it is placed on the ground. When the elevator accelerates downward with a_0 (< g), then -



(A) The limiting friction force between the block M and the surface decreases

(B) The system can accelerate with respect to the elevator even when $m < \mu M$

(C) The system does not accelerate with respect to the elevator unless $m > \mu M$

(D) The tension in the string decreases

Q8. When the downward acceleration of the elevator becomes equal to g, then -

(A) Both the blocks remain stationary with respect to the elevator

(B) Both the blocks accelerate vertically downwards with g with respect to ground

(C) The tension in the string becomes equal to zero

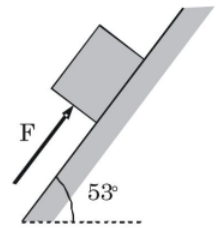
(D) The friction force between the block M and the surface is zero

MATCHING LIST TYPE

*Each question has TWO (02) **matching lists : LIST I and LIST II**. FOUR options are given representing matching of elements from from LIST I and LIST II. ONLY ONE of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.*

Q9. A 50 kg box is held at rest on a slope of inclination 53° above the horizontal. It is released and immediately pushed by a force F up the slope. Coefficients of static and kinetic friction between the slope and the box are 0.5 and 0.4 respectively.

[$g = 10\ m/s^2$]



List-I (Magnitude of force F)	List-II (Friction force)
(P) 50 N	(1) 150 N up the slope
(Q) 150 N	(2) 50 N up the slope
(R) 250 N	(3) 120 N up the slope
(S) 350 N	(4) 400 N up the slope

Code :

	P	Q	R	S
(A)	1	1	2	4
(B)	2	4	3	1
(C)	3	3	1	2
(D)	3	2	1	4

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q10. A block is moving on an inclined plane making an angle of 45° with the horizontal surface and the coefficient of friction is μ. The force required to push up the inclined plane, is 3 times the force required to prevent it from sliding down. If we define $N = 10\ \mu$, then find the value of N.

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

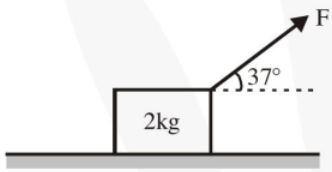
Q1. A block of mass 2 kg is kept on a rough horizontal floor and pulled with a force F. If the coefficient of friction is 0.5. then the minimum force required to move the block is

(A) 10 N

(B) $\frac{100}{11}$ N

(C) $\frac{100}{8}$ N

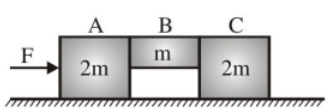
(D) 20 N



Q2. The rear side of a truck is open and a box of mass 20 kg is placed on the truck 4 m away from the open end, $\mu = 0.15$ and $g=10\ m/s^2$. The truck starts from rest with an acceleration of $2m/s^2$ on a straight road. The distance moved by the truck when box starts fall down is

(A) 4 m (B) 8 m (C) 16 m (D) 32 m

Q3. The system is pushed by a force F as shown in figure. All surfaces are smooth except between B and C. Friction coefficient between B and C is μ. Minimum value of F to prevent block B from downward slipping is :-



(A) $\left(\frac{3}{2\mu}\right)mg$ (B) $\left(\frac{5}{2\mu}\right)mg$

(C) $\left(\frac{5}{2}\right)\mu mg$ (D) $\left(\frac{3}{2}\right)\mu mg$

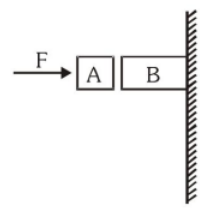
Q4. Given in the figure are two blocks A and B of weight 20 N and 100 N, respectively. These are being pressed against a wall by a force F as shown. If the coefficient of friction between the blocks is 0.1 and between block B and the wall is 0.15, the frictional force applied by the wall on block B is

(A) 120 N

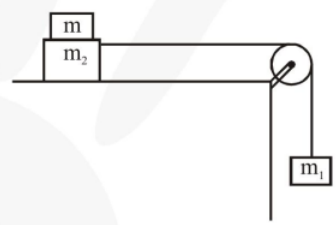
(B) 150 N

(C) 100 N

(D) 80 N



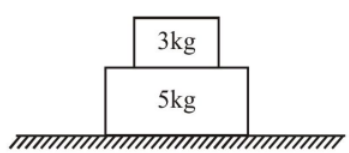
Q5. In figure coefficient of friction between m_2 and the horizontal surface is μ, minimum value of m so that the system has zero acceleration, is :-



(A) $\mu(m_1 + m_2)$ (B) $\mu(m_1 - m_2)$

(C) $m_2 - \frac{m_1}{\mu}$ (D) $\frac{m_1}{\mu} - m_2$

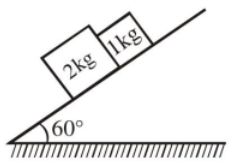
Q6. Two blocks are placed on a horizontal surface as shown in the figure. The coefficient of friction between the blocks is 0.5 and between block of 5 kg and the surface is 0.7. The maximum horizontal force that can be applied to the block of mass 5 kg so that the two blocks move without slipping is : (Take $g = 10\ m/s^2$)



(A) 14 N (B) 24 N

(C) 56 N (D) 96 N

Q7. In the figure shown if friction coefficient of block 1kg and 2kg with inclined plane is μ_1=0.5 and μ_2 = 0.4 respectively, then



(A) both block will move together.

(B) both block will move separately.

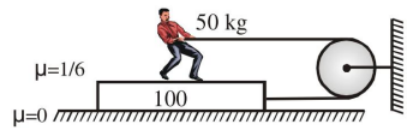
(C) there is a non zero contact force between two blocks.

(D) none of these

Q8. The time taken by a body to slide down a rough 45° inclined plane is twice that required to slide down a smooth 45° inclined plane. The cofficient of kinetic friction between the object and rough plane is given by :-

(A) $\frac{1}{3}$ (B) $\frac{3}{4}$ (C) $\sqrt{\frac{3}{4}}$ (D) $\sqrt{\frac{4}{3}}$

Q9. A man of mass 50 kg is pulling on a plank of mass 100 kg kept on a smooth floor as shown with force of 100 N. If both man & plank move together, find force of friction acting on man.



(A) $\frac{100}{3}$ N towards left

(B) $\frac{100}{3}$ N towards right

(C) $\frac{250}{3}$ N towards left

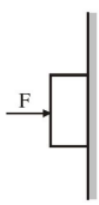
(D) $\frac{250}{3}$ N towards right

Exercise - 2A — *Brain Booster*

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. A block of mass 1 kg is held at rest against a rough vertical surface by pushing by a force F horizontally. The coefficient of friction is 0.5. When



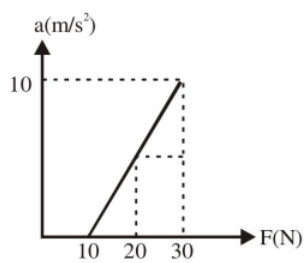
(A) F = 40 N, friction on the block is 20 N.

(B) F = 30 N, friction on the block is 10 N.

(C) F = 20 N, friction on the block is 10 N.

(D) Minimum value of force F to keep block at rest is 20 N.

Q2. A block placed on a rough horizontal surface is pushed with a force F acting horizontally on the block. The magnitude of F is increased and acceleration produced is plotted in the graph shown.



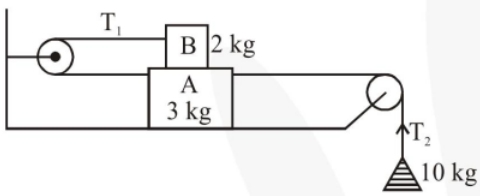
(A) Mass of the block is 2 kg.

(B) Coefficient of friction between block and surface is 0.5.

(C) Limiting friction between block and surface is 10 N.

(D) When $F = 8$ N, friction between block and surface is 10 N.

Q3. In the figure block A and B are of masses 3kg and 2kg. The coefficient of friction between the two blocks A and B is 0.3. The surface of the table is smooth. Then:-



(a) The acceleration of masses is 5.86 ms^{-2}

(b) The tensions are T_1 and T_2 in the strings are 90.36 N and 17.55 N

(c) Acceleration of masses is 8.15 ms^{-2}

(d) Tension T_1 and T_2 in the strings are 17.72 N and 41.4 N

(A) a, c (B) c, d

(C) b, d (D) a, d

Q4. A block is kept on a rough surface and applied with a horizontal force as shown which is gradually increasing from zero. The coefficient of static and kinetic friction are $1/\sqrt{3}$ then



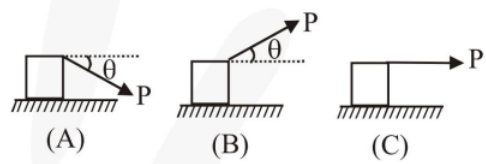
(A) When F is less than the limiting friction, angle made by net force on the block by the surface is less than 30° with vertical.

(B) When the block is just about to move, the angle made by net force by the surface on the block becomes equal to 30° with vertical.

(C) When the block starts to accelerate, the angle made by net force by the surface on the block becomes constant and equal to 30° vertical.

(D) The angle made by net force with vertical on the block by the surface, depends on the mass of the block.

Q5. Figure shows three blocks on a rough surface under influence of a force P of same magnitude in all the three cases. Coefficient of friction is same between each block and ground. What possible relation holds between magnitudes of normal reaction and friction forces. (Assume that blocks do not overturn about edge.) Here, f_A, f_B & f_C are frictional forces & N_A, N_B & N_C are reactions.



(A) $N_A > N_C > N_B$ (B) $f_A > f_C > f_B$

(C) $f_C > f_A = f_B$ (D) $N_C > N_A = N_B$

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Question No. 6 to 10

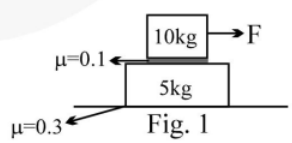


Fig. 1

Q6. When $F = 2$N, the frictional force between 5 kg block and ground is :

(A) 2N (B) 0

(C) 8 N (D) 10 N

Q7. When $F = 2N$, the frictional force between 10 kg block and 5 kg block is :

(A) 2N (B) 15 N

(C) 10 N (D) None

Q8. The maximum F which will not cause motion of any of the blocks is

(A) 10 N (B) 15 N

(C) data insufficient (D) None

Q9. The maximum acceleration of 5 kg block is

(A) 1 m/s^2 (B) 3 m/s^2

(C) 0 (D) None

Q10. The acceleration of 10 kg block when $F = 30$ N is

(A) 2 m/s^2 (B) 3 m/s^2

(C) 1 m/s^2 (D) None

Answer Key

Ex-1											
1	2	3	4	5	6	7	8	9	10	11	12
A	A	A	A	A	A	D	C	A	A	C	A
13	14	15	16	17							
B	C	C	A	D							

Ex- 1A

1. C,D **2.** A,C **3.** B,C **4.** B,C **5.** C

6. A,C **7.** A,C,D **8.** A,B,C,D **9.** C **10.** 5

Ex-2								
1	2	3	4	5	6	7	8	9
B	C	B	A	D	D	B	B	A

Ex- 2A

1. B,C,D **2.** A,B,C **3.** D **4.** A,B,CD **5.** A,B,C

6. A **7.** A **8.** A **9.** C **10.** A

-: Important Notes & Formulas :-